

MECH5200 Numerical Methods for PDEs : *Course Introduction*

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Example PDEs

- PDE: A differential equation involving an unknown function(s) of several variables and their partial derivatives w.r.t those variables.
- Partial differential equations are convenient mathematical *models* of physical system
 - e.g., Indicates how u changes with respect to x, y and/or t ?
- They are found everywhere in engineering and mathematics

Can you give an example of a PDE?

PDE Example: Laplace and Poisson Equations

The Poisson Equation (Elliptic):

$$-\kappa \nabla^2 u = -\kappa \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) = f \quad (1)$$

Laplace's Equation (Elliptic):

$$\nabla^2 u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \quad (2)$$

- Electrostatics and capacitance extraction, Potential flow, Torsion in a bar of constant cross section, Gravity driven viscous flow in a channel, Membrane deflection, Ground water flow, Stationary heat transfer ...
- Any ideas why this partial differential equation is prevalent?
- What does this equation model?
- What are the properties of solutions of this equation?

PDE Example: Laplace and Poisson Equations

- Example: *ArbTwoD_FD.m*

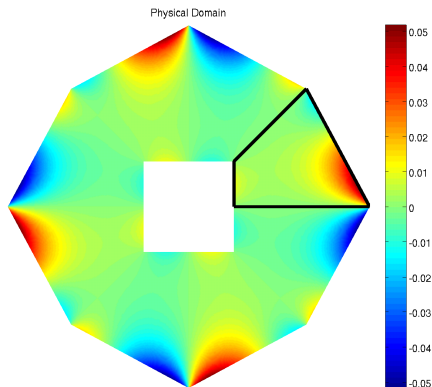


Figure: Torsion in a bar solved using Finite Differences

Laplace Equation: A Fun Example

- The MathWorks logo can be interpreted as a harmonic function
- It satisfies wave's equation:
- See also the MathWorks technical article on related eigenfunction properties:
The MathWorks logo as an eigenfunction of the wave equation

Figure: The MATLAB logo as a solution of Laplace's equation

PDE Example: Unsteady Heat Conduction

- Let's make the equation a bit more complicated and see what happens.
- Unsteady heat conduction equation (Parabolic):

$$\frac{\partial u}{\partial t} = \kappa \nabla^2 u + f \quad (3)$$

$$f = \frac{\partial u}{\partial t} - \kappa \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) \quad (4)$$

- We've simply added a $\frac{\partial u}{\partial t}$ term.
- What does this equation represent/mean?
- How can we interpret this equation's behavior?
- What can we say about the solutions of this equation?

PDE Example: Unsteady Heat Conduction

- Example: *OneD_FD_Temp_EX.m*
- Example: *TempImplicit.m*
- 3blue-1brown:
<https://youtu.be/ly4S0oi3Yz8?feature=shared>

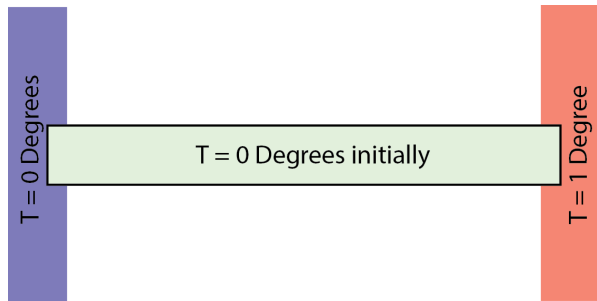


Figure: Model problem for unsteady temperature problem

Frist order PDEs with a new/different behavior

The 1st Order Wave Equation (Advection Equation):

$$\frac{\partial u}{\partial t} + c \cdot \nabla u = 0 \quad (5)$$

or in 1-D:

$$\frac{\partial u}{\partial t} + c \cdot \frac{\partial u}{\partial x} = 0 \quad (6)$$

What does this equation represent?

How can we interpret this equation?

How do the solutions of this equation behave?

First Order Wave Equation (advection equation)

- Example: *HyperWave_EX.m*

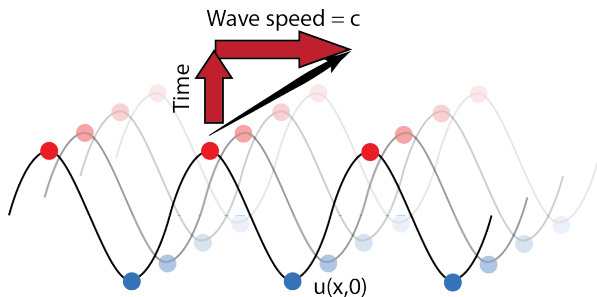


Figure: Model for the wave equation (showing time and space)

Combined Convection-Diffusion Equation

- Combining the diffusion (Laplace operator) and the advection equation together – yields a convection-diffusion model equation (useful for describing transport):

$$\frac{\partial u}{\partial t} + U \cdot \nabla u = \kappa \nabla^2 u + f \quad (7)$$

Where $U, \kappa > 0$, f may be constant or functions in space. U is typically a field velocity, f is a source term and κ is a physical parameter related to the diffusion physics of the problem.

PDEs : The Convection-Diffusion Equation

$$\frac{\partial u}{\partial t} + U \cdot \nabla u = \kappa \nabla^2 u + f \quad (8)$$

- $u = T$ (temperature) – this is the Heat Transfer Equation
- $u = P$ (pollutant concentration) – Coastal/groundwater engineering
- $u = p$ (price of an option) – Financial Engineering
- $u = \vec{u}$ (momentum per unit mass or velocity) – Navier Stokes Equation

Obviously, the convection-diffusion equation is an important PDE.

Some other examples of equations – interesting behavior

Nonlinear KdV equation (Solitons.m):

$$\frac{\partial u}{\partial t} + \frac{\partial^3 u}{\partial x^3} - 6u \frac{\partial u}{\partial x} = 0 \quad (9)$$

- Weakly non-linear, shallow water equation, optics, etc.
- Rate of change with respect to multiple variables (or dimensions).

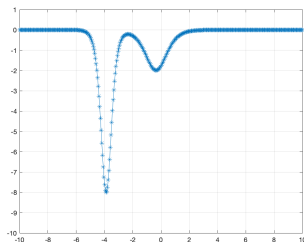


Figure: The KdV equation solution – shows interacting traveling solitons

Compiled vs. Interpreted Coding Languages

Computer programming languages are classified as:

- Interpreted languages (eg. MATLAB, Python, Octave, etc.)
 - Code is saved as written
 - Instructions are sequentially compiled as they are reached
 - Slow(er) run-time
 - Fast to prototype and code (less errors)
 - Usually used for prototyping
- Compiled languages (eg: C/C++/Fortran)
 - Code is converted to machine instructions and saved as an executable
 - Fast run-time → just need to run the instructions
 - Slower to prototype and code (more errors)
 - Usually used for production and released codes

Interpreted Coding Languages

MATLAB is an interpreted language – designed for vector and matrix algebra. Thus, there are ways to make MATLAB interpreted code more efficient by making use of built in compiled functions.

- Many of the matrix-matrix, matrix-vector, and vector-vector operations are compiled.
- Writing vectorized code is much more efficient than loop structures

Try the following:

- Generate a 1000×1000 matrix, A , and multiply it by a vector, b .
- Use for loops in one case and use the $*$ operator in the other case

We can see that MATLABs built-in operations are much better. Try to keep **for**, **while**, etc. loops to a minimum.

Interpreted Coding Languages — MATLAB vs. Python

- **MATLAB** is often simpler to use for purely scientific computing
 - Excellent, centralized documentation tailored to numerical methods
 - Many algorithms and solvers are available out of the box
 - Requires a paid license
- **Python** is free and open-source
 - Scientific computing relies on multiple external libraries
 - Documentation is more fragmented and setup can be more involved
 - Greater flexibility beyond scientific computing
- Choice often depends on whether ease-of-use or cost/flexibility is the priority